Angle Modulation

- Carrier $v_c(t) = V_c \cos(\omega_c t + \phi_c)$
  - V_c = amplitude; angle $\Theta(t) = \omega_c t + \phi_c$
- Angle modulation includes
  - FM and PM
- Difference between FM and PM is
  - how $\Theta(t)$ is varied by $v_m(t)$

Angle Modulation

- For unmodulated carrier

$$\Theta(t) = \omega_c t + \phi_c = \theta_c + \phi_c$$

- if ϕ_c is made proportional to $v_m(t)$, we have PM

$$\Theta(t) = \omega_c t + K_p v_m(t)$$

  - K_p = constant, $\phi = K_p v_m(t)$ = the phase deviation

- If $v_m(t) = V_m \sin \omega_m t$

$$\Theta(t) = \omega_c t + K_p V_m \sin \omega_m t$$

Angle Modulation

- Phase deviation

$$\phi = K_p V_m \sin \omega_m t$$

- Peak phase deviation (in rad) = phase modulation index m_p

$$\phi_{\max} = K_p V_m = m_p$$

$$K_p = \frac{\phi}{V_m} = \text{phase modulator sensitivity (rad/volt)}$$

Angle Modulation

- For FM, f_c is made proportional to $v_m(t)$

$$\omega(t) = \frac{d\Theta(t)}{dt} = \omega_c + K_f v_m(t) \qquad K_f = \text{constant}$$

- Integrating

$$\Theta(t) = \omega_c t + K_f \int_0^t v_m(t)$$

Angle Modulation

- For PM

$$v_c(t) = V_c \cos\left[\omega_c t + K_f v_m(t)\right]$$

- For FM

$$v_c(t) = V_c \cos\left[\omega_c t + K_f \int_0^t v_m(t)\right]$$

- FM is obtained from PM by integrating $v_m(t)$

Frequency Modulation

- If modulating signal is

$$v_m(t) = V_m \cos \omega_m t$$

- For FM

$$v_c(t) = V_c \cos\left[\omega_c t + K_f \int_0^t V_m \cos \omega_m t \, dt \right]$$

$$= V_c \cos\left[\omega_c t + \frac{K_f}{\omega_m} V_m \sin \omega_m t \right]$$

$$= V_c \cos\left[\omega_c t + m \sin \omega_m t\right]$$

Frequency Modulation

- Peak carrier (frequency) deviation is

$$\delta = \frac{K_f V_m}{2\pi}$$

- Modulation index m is

$$m = \frac{\delta}{f_m}$$

Frequency Modulation

- Equation $v_c(t) = V_c \cos[\omega_c t + m \sin \omega_m t]$ can be written as

$$\frac{v_c(t)}{V_c} = J_0(m)\cos\omega_c t + J_1(m)[\cos(\omega_c+\omega_m)t - \cos(\omega_c-\omega_m)t] + J_2(m)[\cos(\omega_c+2\omega_m)t - \cos(\omega_c-2\omega_m)t] + \ldots$$

$J_n(m)$ are Bessel functions of order n of the first kind

- From Bessel table

J_0 – the carrier, J_1 – first pair of side freqs J_2 – 2nd pair of side freqs, etc

Frequency Modulation

- Side frequencies < 1% of unmodulated carrier are not listed in Bessel Table

Frequency Modulation

Bessel Functions of the First Kind, $J_n(x)$

x	J_0	J_1	J_2	J_3	J_4	J_5	J_6	J_7	J_8	J_9	J_{10}
0.0	1.00										
0.2	0.99	0.10									
0.4	0.96	0.20	0.02								
0.6	0.91	0.29	0.04								
0.8	0.85	0.37	0.08	0.01							
1.0	0.77	0.44	0.11	0.02							
1.2	0.67	0.50	0.16	0.03	0.01						
1.4	0.57	0.54	0.21	0.05	0.01						
1.6	0.46	0.57	0.26	0.07	0.01						
1.8	0.34	0.58	0.31	0.10	0.02						
2.0	0.22	0.58	0.35	0.13	0.03	0.01					
2.2	0.11	0.56	0.40	0.16	0.05	0.01					
2.4	0.00	0.52	0.43	0.20	0.06	0.02					
2.6	(0.10)	0.47	0.46	0.24	0.08	0.02	0.01				
2.8	(0.19)	0.41	0.48	0.27	0.11	0.03	0.01				
3.0	(0.26)	0.34	0.49	0.31	0.13	0.04	0.01				
3.2	(0.32)	0.26	0.48	0.34	0.16	0.06	0.02				

Frequency Modulation

3.4	(0.36)	0.18	0.47	0.37	0.19	0.07	0.02	0.01			
3.6	(0.39)	0.10	0.44	0.40	0.22	0.09	0.03	0.01			
3.8	(0.40)	0.01	0.41	0.42	0.25	0.11	0.04	0.01			
4.0	(0.40)	(0.07)	0.36	0.43	0.28	0.13	0.05	0.02			
4.2	(0.38)	(0.14)	0.31	0.43	0.31	0.16	0.06	0.02	0.01		
4.4	(0.34)	(0.20)	0.25	0.43	0.34	0.18	0.08	0.03	0.01		
4.6	(0.30)	(0.26)	0.18	0.42	0.36	0.21	0.09	0.03	0.01		
4.8	(0.24)	(0.30)	0.12	0.40	0.38	0.23	0.11	0.04	0.01		
5.0	(0.18)	(0.33)	0.05	0.36	0.39	0.26	0.13	0.05	0.02	0.01	
5.2	(0.11)	(0.34)	(0.02)	0.33	0.40	0.29	0.15	0.07	0.02	0.01	
5.4	(0.04)	(0.35)	(0.09)	0.28	0.40	0.31	0.18	0.08	0.03	0.01	
5.6	0.03	(0.33)	(0.15)	0.23	0.39	0.33	0.20	0.09	0.04	0.01	
5.8	0.09	(0.31)	(0.20)	0.17	0.38	0.35	0.22	0.11	0.50	0.02	0.01
6.0	0.15	(0.28)	(0.24)	0.11	0.36	0.36	0.25	0.13	0.06	0.02	0.01
6.2	0.20	(0.23)	(0.28)	0.05	0.33	0.37	0.27	0.15	0.07	0.03	0.01
6.4	0.24	(0.18)	(0.30)	(0.01)	0.29	0.37	0.29	0.17	0.08	0.03	0.01
6.6	0.27	(0.12)	(0.31)	(0.06)	0.25	0.37	0.31	0.19	0.10	0.04	0.01
6.8	0.29	(0.07)	(0.31)	(0.12)	0.21	0.36	0.33	0.21	0.11	0.05	0.02

Frequency Modulation

7.0	0.30	0.00	(0.30)	(0.17)	0.12	0.35	0.34	0.23	0.13	0.06	0.02
7.2	0.30	0.05	(0.28)	(0.21)	0.11	0.33	0.35	0.25	0.15	0.07	0.03
7.4	0.28	0.11	(0.25)	(0.24)	0.50	0.30	0.35	0.27	0.16	0.08	0.04
7.6	0.25	0.16	(0.21)	(0.27)	0.00	0.27	0.35	0.29	0.18	0.10	0.04
7.8	0.22	0.20	(0.16)	(0.29)	(0.06)	0.23	0.35	0.31	0.20	0.11	0.05
8.0	0.17	0.23	(0.11)	(0.29)	(0.11)	0.19	0.34	0.32	0.22	0.13	0.06
8.2	0.12	0.26	(0.06)	(0.29)	(0.15)	0.14	0.32	0.33	0.24	0.14	0.07
8.4	0.07	0.27	0.00	(0.27)	(0.19)	0.09	0.30	0.34	0.26	0.16	0.08
8.6	0.01	0.27	0.05	(0.25)	(0.22)	0.04	0.27	0.34	0.28	0.18	0.10
8.8	(0.04)	0.26	0.10	(0.22)	(0.25)	(0.01)	0.24	0.34	0.29	0.20	0.11
9.0	(0.09)	0.25	0.14	(0.18)	(0.27)	(0.06)	0.20	0.33	0.31	0.21	0.12
9.2	(0.14)	0.22	0.18	(0.14)	(0.27)	(0.10)	0.16	0.31	0.31	0.23	0.14
9.4	(0.18)	0.18	0.22	(0.09)	(0.27)	(0.14)	0.12	0.30	0.32	0.25	0.16
9.6	(0.21)	0.14	0.24	(0.04)	(0.26)	(0.18)	0.08	0.27	0.32	0.27	0.17
9.8	(0.23)	0.09	0.25	(0.01)	(0.25)	(0.21)	0.03	0.25	0.32	0.28	0.19
10.0	(0.25)	0.04	0.25	0.06	(0.22)	(0.23)	(0.01)	0.22	0.32	0.29	0.21

NB: Numbers in bracket indicate negatives (e.g -0.27 = (0.27))